10 Understanding Computer Vision for Intelligent Autonomous Systems

Summiya Parveen[1]* and Aruna Tomar[2]

[1]*Department of Mathematics, College of Allied Sciences, COER University, Roorkee, Rehmadpur, Uttarakhand, India*

[2]*Department of Physics, College of Allied Sciences, COER University, Roorkee, Rehmadpur, Uttarakhand, India*

Abstract

Computer vision is a field of artificial intelligence that focuses on enabling machines to interpret and understand visual data from the world around them. Intelligent autonomous system (IAS) play a crucial role in

automating tasks that require visual perception, such as object recognition, image classification, and scene understanding.

At a high level, computer vision works by using algorithms and mathematical models to analyze and extract information from digital images or videos. This process involves several steps, including image acquisition, pre-processing, feature extraction, and classification.

Image acquisition involves capturing visual data using cameras or other sensors. Pre-processing involves cleaning and enhancing the image to improve its quality and make it easier to analyze. Feature extraction involves identifying key visual features in the image, such as edges, corners, and textures. Finally, classification involves using machine learning algorithms to classify the image based on its features and assign it to a specific category or label.

In IAS, computer vision automates tasks such as quality control in manufacturing, object recognition in logistics, and facial recognition in security systems. Leveraging computer vision enhances efficiency, accuracy, and safety across various industries and applications.

Keywords**:** Computer vision, intelligent autonomous systems, image processing, deep learning, multi-modal fusion

10.1 Introduction

The emergence of intelligent autonomous systems (IAS) has marked a new era of technological advancement, transforming our understanding and interaction with the environment. The key to these sophisticated systems is the field of computer vision, which is a vital branch of artificial intelligence that enables machines to process and analyze visual data from the real world [1, 2]. This field is crucial in empowering IAS to perceive and comprehend their surroundings, which serves as a foundation for a broad spectrum of applications across numerous sectors. Extracting valuable information from digital images and videos is a fundamental aspect of computer vision. This intricate process encompasses a series of discrete steps, such as image acquisition, pre-processing, feature extraction,

and classification [3]. The recent advancements in deep learning, an influential subfield of machine learning, have dramatically transformed the field of computer vision.

The potent capacity of deep neural networks to recognize intricate patterns and representations from copious visual data has facilitated the creation of exceptionally sophisticated computer vision models [4, 5].

Undoubtedly, the rapid progress in the field of computer vision has led to remarkable advancements in object detection, image classification, and semantic segmentation tasks. Pivotal research studies, including the introduction of convolutional neural networks for large-scale image classification on the ImageNet dataset [6], the formulation of regionbased object detection frameworks such as Faster R-CNN [16], and the emergence of highly effective semantic segmentation models like Mask R-CNN [8], have significantly contributed to the current state-of-the-art in computer vision systems.

The progress of computer vision and intelligent autonomous systems has been significantly accelerated by access to extensive, varied datasets, including ImageNet [19], COCO [17], and Pascal VOC [18]. These annotated resources have empowered researchers to develop more sophisticated and broadly applicable models capable of dealing with intricate real-world situations.

The growth and influence of these technologies are apparent from the expanding applications across diverse sectors, as depicted in Figure 10.1.

The following line chart reveals the progress of the market size (in billion USD) for computer vision and intelligent autonomous system (IAS) applications across five key sectors, namely, manufacturing, logistics, healthcare, security, and retail, from 2018 to 2022. The data on display showcases the consistent rise in the adoption and penetration of these technologies over the past five years, emphasizing the growing significance of computer vision and IAS in transforming various industries. By harnessing these capabilities, businesses can optimize efficiency, boost decision-making, and drive innovation across different sectors.

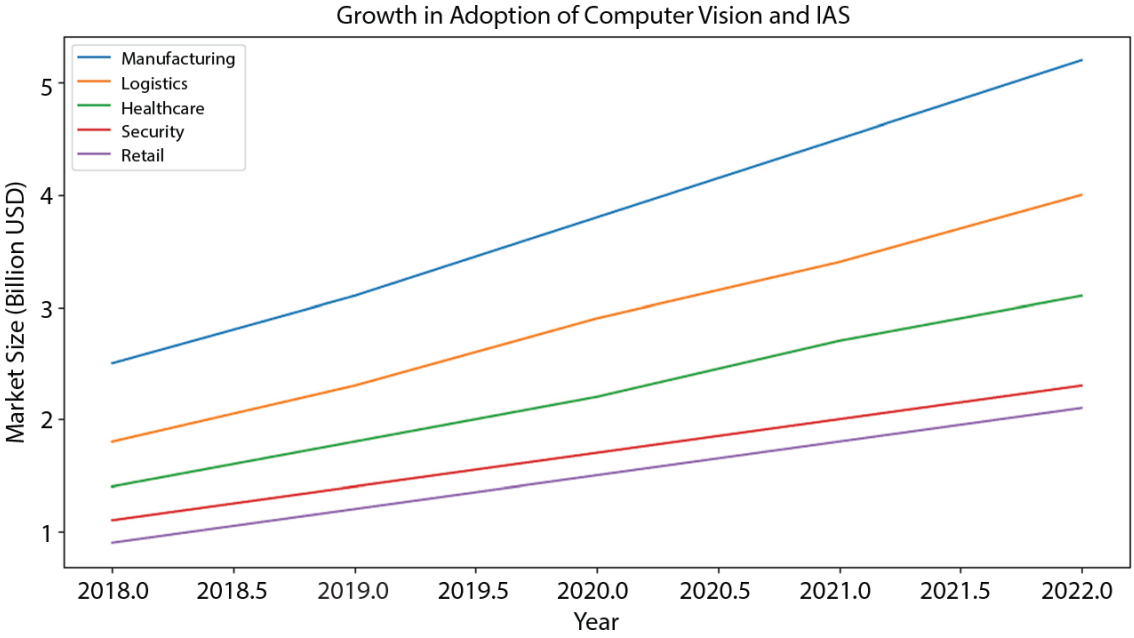


Figure 10.1 Growth in adoption of computer vision and IAS applications across different industries.

In the context of IAS, computer vision plays a crucial role in automating tasks that require visual perception and comprehension. Its applications span a broad range of domains, such as quality control in manufacturing processes [15], object recognition and scene understanding for autonomous vehicles and robotics [21], facial recognition and surveillance for security systems [14], and medical imaging analysis and surgical robotics in healthcare [25]. With the help of computer vision, IAS can enhance efficiency, precision, and safety in a multitude of industries and applications.

However, despite the remarkable progress made in computer vision, several challenges remain to be addressed. Handling complex visual scenes with occlusions, varying lighting conditions, and cluttered environments [22] achieving real-time performance for time-critical applications [23] overcoming data scarcity and diversity issues, and ensuring robustness and generalization to unseen data and adversarial attacks are some of the key challenges that researchers continue to grapple with.

As research in computer vision continues to advance, addressing these challenges and leveraging emerging techniques such as attention mechanisms [14], transformer architectures [15], and novel network architectures [27–29], the capabilities of IAS will be further enhanced, ushering in a new era of intelligent automation across various sectors. This paper

aims to provide a comprehensive understanding of computer vision in the context of IAS, exploring its fundamental principles, key developments, applications, future directions, and the synergistic relationship between these two rapidly evolving fields.

10.2 Fundamentals of Computer Vision

Computer vision is a multidisciplinary field that combines principles from computer science, mathematics, and signal processing to enable machines to interpret and analyze visual data from the real world. This section delves into the fundamental concepts and steps involved in the computer vision pipeline.

10.2.1 Image Acquisition

The first step in the computer vision process is acquiring visual data from the environment. This is typically accomplished through the use of cameras and sensors, which capture digital images or videos. Different types of cameras and sensors may be employed depending on the specific application and requirements, such as visible light cameras, infrared cameras, depth sensors, or multispectral imaging devices.

Cameras and sensors cameras are the most common type of visual sensor used in computer vision applications. They capture light and convert it into digital signals that can be processed by computers. There are various types of cameras, each with its own characteristics and capabilities:

- Visible light cameras: These capture images in the visible spectrum, similar to how the human eye perceives the world. They are widely used in a variety of applications, from consumer photography to industrial quality control.
- Infrared (IR) cameras: These cameras detect infrared radiation, which is invisible to the human eye but can provide valuable information about temperature, heat signatures, and other thermal properties. IR cameras are often used in security, surveillance, and night vision applications.

- Depth sensors: These specialized sensors, such as time-of-flight (ToF) cameras or structured light sensors, can measure the distance between the camera and objects in the scene. Depth information is crucial for applications like 3D reconstruction, robot navigation, and augmented reality.
- Multispectral and hyperspectral cameras: These sensors capture images across multiple wavelength bands, beyond the visible spectrum. They can provide additional information about the chemical composition, material properties, and other characteristics of the objects being imaged. These cameras are commonly used in remote sensing, agriculture, and medical imaging applications.

The type of visual data acquired can vary widely, depending on the application and the sensors used. The most common types of visual data in computer vision include:

- 2D images: These are the most common form of visual data, consisting of a grid of pixels representing the intensity or color values at each location in the image.
- Video sequences: Video data is a series of temporally related 2D images, capturing the dynamic changes in a scene over time.
- 3D point clouds: These are collections of 3D coordinate points, often obtained from depth sensors or laser scanners, representing the spatial structure of a scene.
- Multispectral and hyperspectral data: This type of visual data consists of multiple image channels, each capturing a different wavelength band, providing additional spectral information about the scene.

The choice of visual data and acquisition method depends on the specific requirements of the computer vision application, such as the level of detail needed, the operating environment, and the computational resources available.

10.2.2 Pre-Processing

Once the visual data is acquired, pre-processing techniques are applied to enhance the data quality and prepare it for further analysis. This stage of-

ten involves image enhancement techniques to improve the visibility of relevant features and reduce noise or artifacts, as well as noise reduction methods to address various sources of unwanted signals in the data.

Image enhancement is the process of improving the interpretability or perceived quality of digital images for a specific application or task. This can involve a variety of techniques, such as:

- Contrast adjustment: Adjusting the contrast of an image can help highlight relevant features and improve the visibility of details. Techniques like histogram equalization, adaptive histogram equalization, and contrast limited adaptive histogram equalization (CLAHE) are commonly used for this purpose.
- Sharpening: Sharpening algorithms, such as the Laplacian filter or unsharp masking, can help enhance the edges and fine details in an image, making it appear more crisp and focused.
- Color correction: Color correction methods, including white balance adjustment, color normalization, and color space transformations, can help correct color-related issues and ensure consistent color representation across different images or sensors.
- DE blurring: Techniques like Wiener filtering or Lucy-Richardson deconvolution can be used to reduce blur in images caused by factors like camera motion, defocus, or atmospheric disturbances.

Noise Reduction Real-world visual data can be affected by various sources of noise, such as sensor imperfections, environmental conditions, or compression artifacts. Noise reduction is an important aspect of preprocessing, as it can significantly improve the quality and reliability of the data for subsequent analysis.

Some common noise reduction techniques include:

- Gaussian filtering: Applying a Gaussian smoothing filter can help reduce high-frequency noise while preserving important image details.
- Median filtering: Median filtering is effective in removing impulse noise, such as salt-and-pepper noise, without blurring edges.
- Wavelet-based de-noising: Wavelet-based methods, like the Wiener filter in the wavelet domain, can adaptively remove noise while preserv-

ing important image structures.
- Non-local means: This technique exploits the self-similarity of image patches to de-noise the image while preserving edges and textures.

The choice of pre-processing techniques and their specific parameters depends on the type of visual data, the nature of the noise or artifacts present, and the requirements of the computer vision application.

10.2.3 Feature Extraction

Feature extraction is a critical step in the computer vision pipeline, where relevant visual features are identified and extracted from the pre-processed data. These features can include edges, corners, textures, shapes, and other salient characteristics that are useful for subsequent analysis and classification tasks.

Edge detection is the process of identifying and locating sharp discontinuities in an image, which often correspond to the boundaries of objects or regions. Edge detection algorithms are fundamental to many computer vision applications, as they provide valuable information about the structure and shape of objects in the scene.

Some commonly used edge detection techniques include:

- Canny edge detector [30]: The Canny edge detector is a widely used algorithm that employs a multi-stage process to detect a wide range of edges, including weak edges, while suppressing noise.

For example, in object detection, edge detection can help identify the edges of objects in an image, which can then be used to accurately localize and recognize the objects. Edge detection is an essential step in lane detection and modeling. Source: Lane Detection and Modeling Using Edge-Based Techniques

Edge detection is a critical step in computer vision tasks such as object detection, image segmentation, and lane detection. It involves identifying boundaries or edges of objects within an image by detecting changes in

pixel intensity or color. You can explore some practical examples of edge detection in Figure 10.2.

- Sobel operator [31]: The Sobel operator computes an approximation of the gradient of the image intensity function, highlighting regions with high spatial frequency that often correspond to edges.
- Corner detection is the process of identifying significant, well-defined points in an image that are invariant to rotation, scale, and other transformations. These corner points, or key points, can be used for a variety of applications, such as object recognition, image registration, and camera calibration.
- Prewitt operator [31]: Similar to the Sobel operator, the Prewitt operator uses a pair of 3x3 convolution masks to estimate the gradient of the image, with a slightly different kernel design.

The Harris corner detector [32] is a popular algorithm that identifies corners by analyzing the local autocorrelation of the image. Other corner detection methods include the FAST (features from accelerated segment test) algorithm and the SIFT (scale-invariant feature transform) keypoint detector.

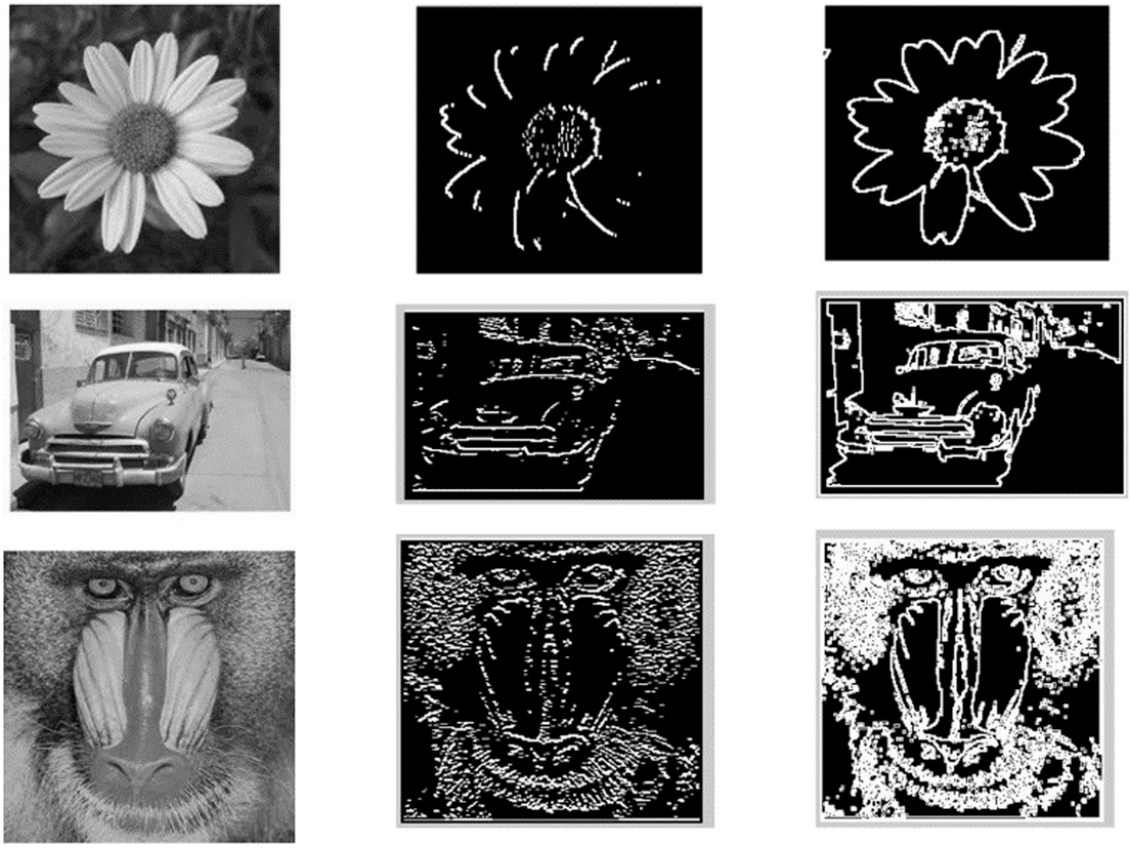


Figure 10.2 Examples of edge detection.

Texture analysis involves the characterization and description of the visual patterns, structures, and relationships within an image. Texture features can be useful for tasks such as image classification, segmentation, and retrieval, as they provide information about the surface properties and material composition of objects in the scene.

Techniques for texture analysis include:

- Statistical methods: These analyze the distribution of intensity values within a local neighborhood, using measures like contrast, correlation, energy, and homogeneity [33].
- Structural methods: These focus on identifying the repeated patterns or primitives that make up the texture, using tools like Markov random fields or morphological operators [33].
- Filter-based methods: These apply a bank of filters, such as Gabor filters or wavelet transforms, to the image and analyze the responses to extract texture features [33, 55].

10.2.4 Classification

The final stage of the computer vision pipeline involves classifying or categorizing the extracted features using machine learning algorithms. This step aims to assign semantic labels or categories to the visual data based on the identified features and patterns.

Machine Learning Algorithms: Various machine learning techniques are employed for image classification and categorization in computer vision. These include both traditional methods and more recent deep learning approaches:

- Traditional methods:
  - Support Vector Machines (SVMs) [34]: SVMs are a popular and effective algorithm for image classification, capable of handling high-dimensional feature spaces.
  - Decision Trees: Decision tree-based models, such as random forests, can be used to build robust and interpretable classifiers for visual data.
  - K-Nearest Neighbors (KNN): The KNN algorithm can be applied to classify images based on the similarity of their extracted features to a labeled training set.
- Deep Learning Methods:
  - Convolutional Neural Networks (CNNs) [6, 11, 12]: CNNs are a class of deep neural networks that have revolutionized image classification and recognition tasks, leveraging the hierarchical extraction of visual features.
  - Recurrent Neural Networks (RNNs): RNNs, such as long short-term memory (LSTMs), can be used for tasks like image captioning and visual question answering, where sequential processing of visual data is required.
  - Transformer Architectures [14, 15]: Recent transformer-based models, like vision transformers, have shown promising results in image classification, outperforming CNN-based approaches in certain scenarios.

Image Categorization: The output of the classification stage can take various forms, depending on the specific computer vision application and the

desired level of semantic understanding. Some common image categorization tasks include:

- Object recognition: Identifying and localizing specific objects within an image, such as cars, pedestrians, or household items.
- Scene understanding: Classifying the overall context or type of scene depicted in an image, such as indoor, outdoor, urban, or natural environments.
- Semantic segmentation: Partitioning an image into meaningful regions or segments, and assigning a semantic label to each segment, such as sky, road, building, or vegetation.
- Image retrieval: Matching and retrieving visually similar images from a database based on their content and features.

10.3 Applications of Computer Vision in IAS

Computer vision plays a crucial role in enabling intelligent autonomous systems to perceive and understand their environment, facilitating a wide range of applications across various industries. This section explores some key areas where computer vision is being leveraged to drive intelligent automation and enhance the capabilities of IAS.

10.3.1 Quality Control in Manufacturing

In the manufacturing industry, computer vision systems play a crucial role in automating quality inspection processes, enabling intelligent autonomous systems (IAS) to detect defects, irregularities, and deviations from specifications with high accuracy and speed. By leveraging advanced computer vision techniques, IAS can perform real-time quality checks, ensuring the consistent production of high-quality goods while reducing the need for manual inspections.

One of the key applications of computer vision in manufacturing quality control is object recognition and defect detection. IAS can be trained to identify various types of defects, such as cracks, scratches, discoloration, or dimensional inaccuracies, by using deep learning-based object detection algorithms like faster R-CNN [16] and YOLO [26]. These models are

able to localize and classify different types of defects within the manufactured products, providing precise feedback to the production system.

In addition to object recognition, computer vision-enabled IAS can also perform imagebased anomaly detection to identify irregular patterns or deviations from expected product characteristics. Techniques like one-class support vector machines (OC-SVMs) [34] and deep auto encoder models [15, 22] can be trained to learn the normal appearance of high-quality products and then flag any instances that deviate significantly from this learned representation.

Furthermore, computer vision can be utilized for automated visual inspection and process monitoring in manufacturing environments. IAS can be equipped with high-resolution cameras and specialized lighting to capture detailed images of the products at various stages of the production line. These images can then be analyzed using computer vision algorithms to assess dimensional accuracy, surface quality, and assembly integrity, enabling real-time feedback and process optimization.

By integrating computer vision into their quality control processes, manufacturing companies can benefit from several advantages offered by IAS:

- Improved inspection accuracy and consistency: Computer vision-based systems can identify defects and anomalies with greater precision and repeatability than manual human inspections.
- Increased production efficiency: Automated quality control reduces the need for manual intervention, allowing for faster processing and reduced production bottlenecks.
- Cost savings: The elimination of manual inspection tasks and the reduction of product waste due to defects can lead to significant cost savings for manufacturing operations.
- Scalability and adaptability: IAS-based quality control solutions can be easily scaled to accommodate changes in production volume and can be adapted to different product lines or manufacturing processes.

As computer vision continues to advance, with improvements in deep learning algorithms, sensor technologies, and computational power, the

integration of these capabilities into intelligent autonomous systems will further enhance the quality, efficiency, and competitiveness of modern manufacturing operations.

10.3.2 Object Recognition in Logistics

Computer vision's ability to recognize and localize objects is invaluable in the logistics and transportation sector. Intelligent autonomous systems (IAS) equipped with computer vision can automate a wide range of tasks in the logistics domain, including inventory management, package sorting, and autonomous navigation, by identifying and tracking the movement of various items and packages throughout the supply chain.

Object recognition techniques, such as region-based convolutional neural networks (R-CNNs) [16] and single-shot detectors (SSDs) [27], enable IAS to accurately detect and classify different types of packages, containers, and other logistics-related objects. This allows for automated inventory tracking, where IAS can constantly monitor the location and status of goods, improving efficiency and reducing errors.

In addition, computer vision-based object recognition can be used for automated package sorting and routing in logistics hubs and warehouses. By identifying the unique characteristics and labels of individual packages, IAS can autonomously sort and direct packages to their correct destinations, streamlining the material handling and distribution processes.

Furthermore, object recognition capabilities are crucial for the development of autonomous logistics vehicles and robots. IAS equipped with computer vision can navigate through complex environments, detect and avoid obstacles, and precisely manoeuver to pick up, transport, and deliver goods, leveraging techniques like semantic segmentation [21, 23, 24] and depth estimation [30].

By integrating computer vision-powered object recognition, logistics operations can benefit from increased efficiency, reduced labor costs, and enhanced overall productivity, as IAS takes on more of the repetitive and labor-intensive tasks in the supply chain.

10.3.3 Facial Recognition in Security Systems

Computer vision-enabled facial recognition is a critical component of intelligent security and surveillance systems. Intelligent autonomous systems (IAS) can use this technology to identify and track individuals, monitor access control, and detect anomalies or suspicious activities, contributing to enhanced public safety and crime prevention efforts.

Facial recognition algorithms such as those based on convolutional neural networks [14] leverage computer vision techniques to extract and match distinctive facial features from captured images or video streams. IAS can use these algorithms to accurately identify and verify the identities of individuals, enabling access control systems, surveillance monitoring, and criminal investigation support.

Beyond simple identification, computer vision-based facial recognition in IAS can also be used for behavioral analysis and anomaly detection. By analyzing the facial expressions, movements, and interactions of individuals, IAS can detect potentially suspicious or concerning behaviors, alerting security personnel or triggering appropriate response protocols.

Additionally, computer vision techniques like object tracking [32] can be integrated with facial recognition to enable the continuous monitoring and surveillance of individuals of interest, even as they move through complex environments.

The deployment of computer vision-powered facial recognition in intelligent autonomous security systems has the potential to enhance public safety, improve the efficiency of security operations, and aid law enforcement in their efforts to prevent and investigate criminal activities. However, the use of these technologies also raises important ethical and privacy considerations that need to be carefully addressed.

10.3.4 Autonomous Vehicles

One of the most prominent applications of computer vision in IAS is autonomous vehicle technology. By combining computer vision techniques like object detection [16, 26, 27], semantic segmentation [21, 23, 24], and

depth estimation [30], autonomous vehicles can perceive their surroundings, identify obstacles, and navigate safely through complex environments, paving the way for the development of self-driving cars and other autonomous transportation systems.

Object Detection and Classification: Object detection algorithms, such as faster R-CNN [16] and YOLO [26, 27] enable autonomous vehicles to accurately detect and classify objects on the road, including other vehicles, pedestrians, traffic signs, and potential hazards. This information is crucial for safe navigation and decision-making.

The proposed model utilized in autonomous cars' object identification process is shown in Figure 10.3. Collecting real-time footage shot by nearby drivers is the first step in the process. Under the driver's video viewer, the real-time videos are then shown. Features from the videos are extracted using the hybrid object detection model. Using their distinct characteristics, the model recognizes various things. Next, pixel values are used to detect the region of interest (RoI) after it has been triggered. Using the retrieved features, the objects are finally categorized based on their type.

Semantic Segmentation and Scene Understanding: Semantic segmentation techniques [21, 23, 24] are used to partition the visual scene into different classes, such as roads, buildings, vegetation, and other environmental elements. This contextual understanding is essential for autonomous vehicles to plan their routes and manoeuver appropriately.

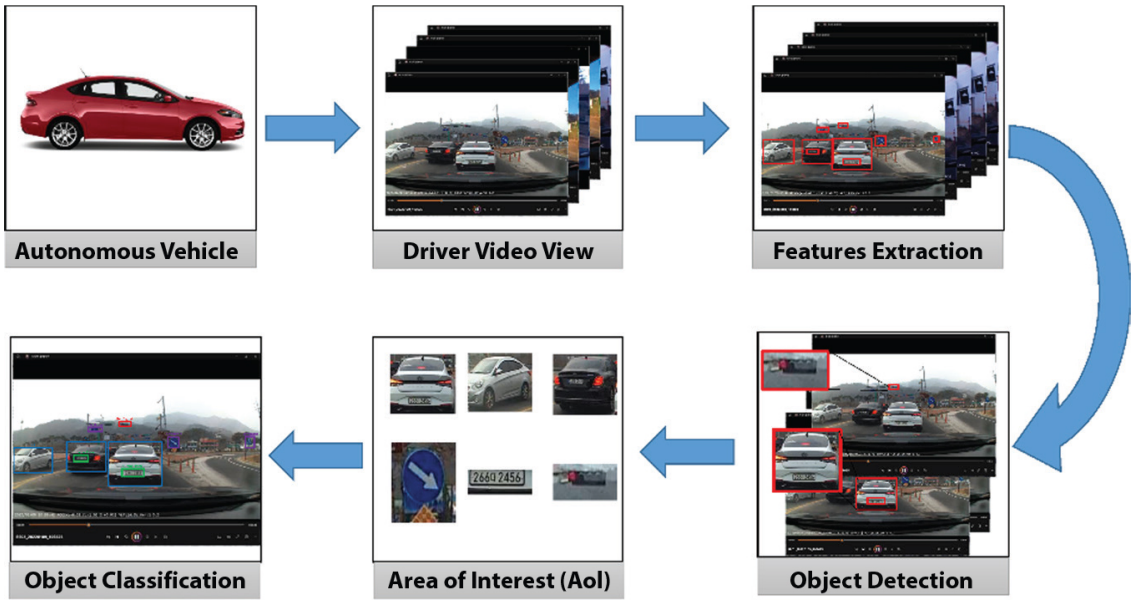


Figure 10.3 The proposed hybrid model for autonomous cars includes an object detecting technique.

Every pixel in a group in Figure 10.4 represents the entire object class. These classes might depict a street scene—traffic signs, cards, people walking, street lamps, or sidewalks—that is significant for autonomous cars.

Depth Estimation and 3D Reconstruction: Depth estimation [30] and 3D reconstruction capabilities, often based on stereo vision or LiDAR sensors, provide autonomous vehicles with a precise understanding of the spatial relationships and distances between objects in their environment, enabling them to navigate safely and avoid collisions.



Figure 10.4 Semantic segmentation of a road scene for an autonomous vehicle.

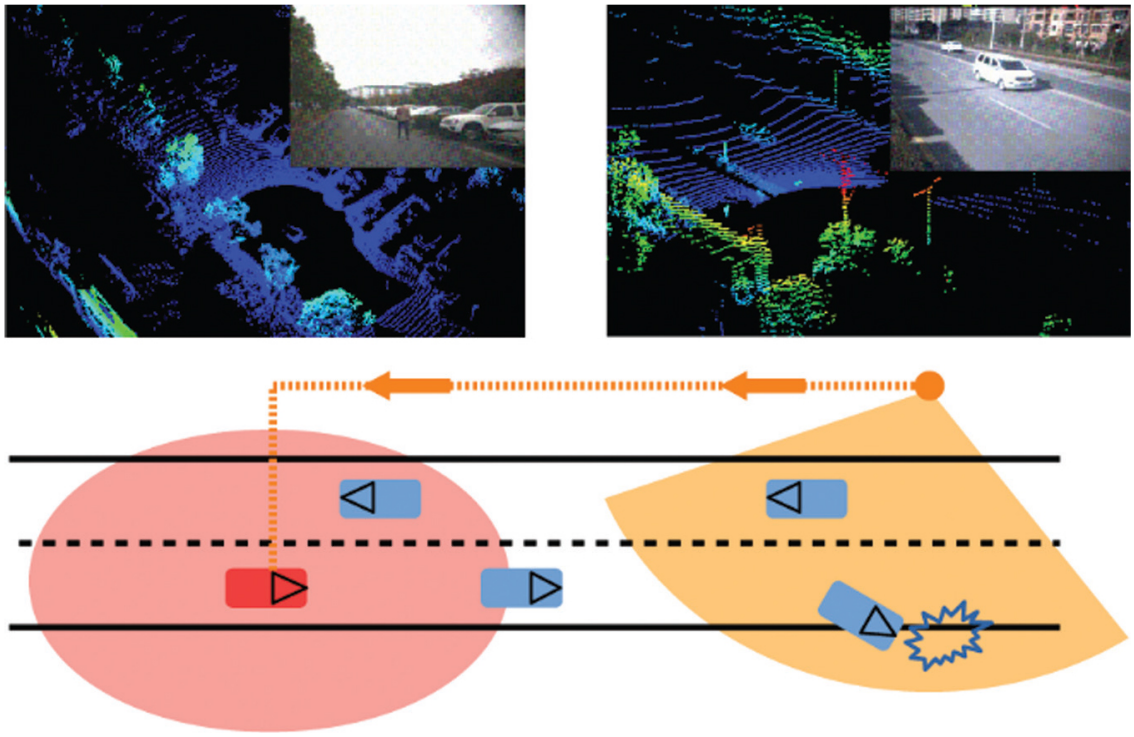


Figure 10.5 3D point cloud representation of the environment for an autonomous vehicle.

To see a picture of how an autonomous car views its surroundings, you can look at Figure 10.5. It shows a special kind of image called a point cloud. Point clouds are made by sensors like LiDAR and help the car understand what's around it in 3D. They're really important for things like spotting objects and understanding the road, which keeps the car driving safely and smoothly.

Object Tracking and Motion Prediction: Computer vision-based object tracking [32] and motion estimation algorithms allow autonomous vehicles to predict the trajectories of dynamic objects, such as other vehicles or pedestrians, enabling proactive decision-making and collision avoidance.

The integration of these computer vision capabilities into IAS is crucial for the development of fully autonomous vehicles, enhancing their situational awareness, decision-making capabilities, and overall safety in complex driving environments.

10.3.5 Medical Imaging

Medical Imaging in the healthcare industry, computer vision is revolutionizing medical imaging analysis. Intelligent autonomous systems (IAS) can assist clinicians by automating tasks like lesion detection, tumor segmentation, and disease diagnosis, using advanced techniques like deep learning-based image classification and segmentation [25, 32].

Image Classification and Anomaly Detection: Computer vision algorithms can be trained on large datasets of medical images, such as X-rays, MRI scans, and CT scans, to learn and recognize the visual patterns associated with various medical conditions or abnormalities. Deep learning models, particularly convolutional neural networks (CNNs) [6, 11, 12] have shown remarkable performance in medical image analysis tasks.

A CNN's design helps it learn different levels of features. The early layers find simple details like texture and shape, while deeper layers figure out more complex things. This helps CNN accurately identify important parts of an image. Since CNNs work with images, they're structured to handle them well. They have layers arranged in three dimensions: height, width, and depth. Figure 10.6 shows the basic setup for image recognition using a CNN. It includes layers like input, convolutional, RELU, pooling, fully connected, and output. These layers work together to make the whole CNN. The convolutional, RELU, and pooling layers extract features, while the fully connected layer handles classification. Understanding how these

layers connect in a deep neural network is trickier than using a pre-made setup.

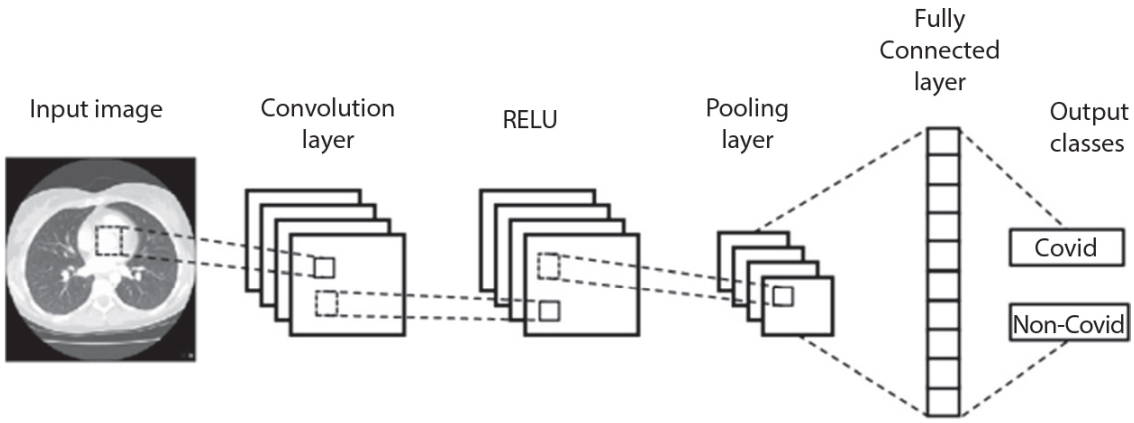


Figure 10.6 A CNN architecture for medical image classification.

Creating a neural network that goes from start to finish can help it understand how different parts work together. By ignoring what doesn't matter, the model gets better at finding important patterns and does the job more effectively. In medicine, neural networks are used for various tasks like sorting, finding, outlining, pinpointing, and matching images. They can help spot COVID-19 earlier, especially when they learn from previous knowledge. This method, called transfer learning, lets us use existing neural networks to analyze x-rays and CT scans of COVID-19 patients.

Object Detection and Localization: IAS can be employed to automatically detect and localize tumors or lesions in medical images, using object detection techniques [16, 26, 27]. This can aid in early diagnosis and treatment planning.

For example: Let us say we have a chest X-ray image of a patient. A neural network trained for lesion detection and localization is fed this image. The network analyzes the image pixel by pixel, looking for areas that deviate from normal lung tissue patterns.

After processing the image, the neural network identifies a suspicious area in the upper right corner of the lung. This area shows characteristics consistent with a tumor.

Next, the network localizes the lesion by providing coordinates that pinpoint its exact location within the image. This information is crucial for

further analysis and treatment planning. With this data, radiologists can review the suspicious area highlighted by the neural network. They can then confirm the presence of the tumor and proceed with appropriate diagnostic tests or treatment options for the patient.

In Figure 10.7, lesion detection and localization algorithms like this assist healthcare professional in identifying abnormalities in medical images, leading to earlier detection and improved patient care.

Semantic Segmentation and Delineation: Semantic segmentation models [23–25] can precisely delineate the boundaries and extent of tumors or other regions of interest, aiding in treatment planning and surgical guidance.

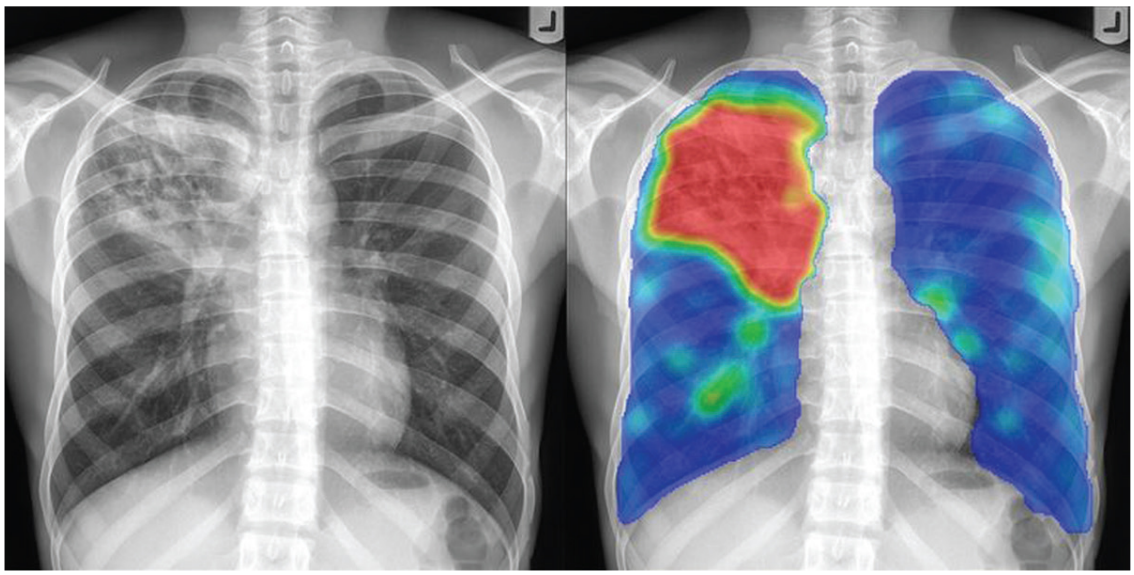


Figure 10.7 Chest x-ray image analysis.

For example: Let us consider an MRI scan of a patient's brain. The scan captures detailed images of the brain's internal structures, including any abnormal growths such as tumors.

A neural network designed for tumor segmentation in MRI scans is tasked with analyzing this image. The network processes each slice of the MRI scan, pixel by pixel, to identify regions that likely belong to the tumor [52, 54].

As the neural network analyzes the MRI scan, it distinguishes between different types of tissues based on their intensity and texture. By compar-

ing these characteristics to a training dataset, the network identifies areas that correspond to the tumor.

Once the tumor is identified, the neural network proceeds to segment it by outlining its boundaries. This segmentation process provides a precise map of the tumor's location within the brain. The segmented tumor region can then be visualized separately, allowing healthcare professionals to assess its size, shape, and position relative to surrounding structures. This information is crucial for treatment planning, such as surgical intervention or targeted radiation therapy. In Figure 10.8, tumor segmentation in MRI scans using neural networks enables accurate detection and localization of tumors, facilitating timely medical intervention and improving patient outcomes.

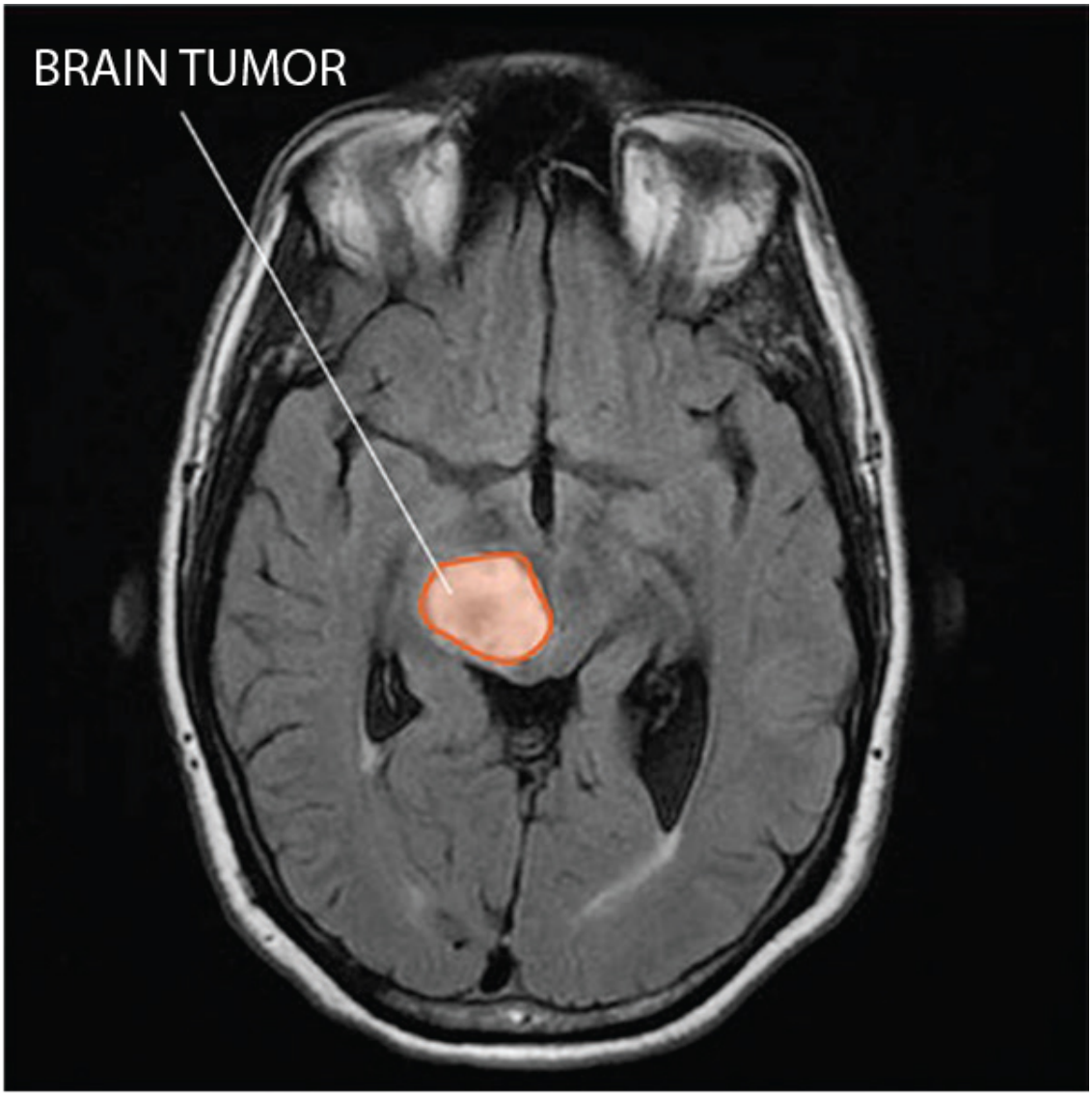


Figure 10.8 Tumor segmentation in an MRI scan.

Multimodal Analysis and Fusion: Computer vision techniques can be combined with other medical imaging modalities, such as functional MRI or positron emission tomography (PET), to provide a more comprehensive understanding of the underlying physiological processes and enhance the diagnostic capabilities of IAS.

For example: Let us consider a patient suspected of having brain cancer. To assist in diagnosis, both MRI and PET scans are performed. The MRI scan provides detailed anatomical images of the brain's structure, while the PET scan offers functional information about metabolic activity.

A multimodal fusion approach combines information from both scans to improve diagnostic accuracy. A neural network designed for multimodal fusion is trained using a dataset that includes paired MRI and PET scans, along with corresponding diagnoses.

During inference, the neural network processes the MRI and PET scans simultaneously, extracting features from each modality. These features capture both structural and functional aspects of the brain, providing a comprehensive view of the disease.

The network then integrates the extracted features using fusion techniques, such as concatenation or attention mechanisms. This fusion process combines complementary information from the MRI and PET scans, enhancing the overall diagnostic performance.

Finally, based on the fused features, the neural network generates a diagnosis or a probability score indicating the likelihood of cancer presence. Healthcare professionals can use this information to make informed decisions about treatment planning and patient management.

By leveraging both MRI and PET data through multimodal fusion, this approach offers a more comprehensive assessment of cancer presence and characteristics, leading to improved diagnostic accuracy and patient care.

With the power of computer vision and deep learning, IAS has the potential to revolutionize medical imaging analysis, streamlining clinical workflows, improving diagnostic accuracy, and ultimately contributing to better patient care and healthcare outcomes.

10.4 Challenges and Emerging Techniques

Despite the remarkable progress made in computer vision and its applications in intelligent autonomous systems (IAS), several challenges remain to be addressed. This section explores two key challenges – robustness to variability and real-time processing – and discusses emerging techniques and approaches aimed at addressing these challenges.

10.4.1 Robustness to Variability

One of the significant challenges in computer vision is ensuring that the algorithms and models are robust to the variability and complexity present in real-world environments. This variability can arise from various factors, such as lighting conditions, occlusions, viewpoint changes,

and the diversity of objects and scenes encountered in different applications.

Lack of robustness can lead to degraded performance and potential failures in computer vision systems, which can have severe consequences in mission-critical applications like autonomous vehicles or medical imaging analysis. To address this challenge, researchers are exploring several techniques:

- Data Augmentation: Data augmentation techniques, such as geometric transformations (rotation, scaling, flipping), color jittering, and synthetic data generation, can be used to artificially expand and diversify the training data, improving the robustness of computer vision models to variations in the input data [35, 36, 51, 55].
- Adversarial Training: Adversarial training methods involve deliberately introducing adversarial examples or perturbations to the training data, forcing the models to learn more robust representations and improve their generalization capabilities [37, 38].
- Domain Adaptation: Domain adaptation techniques aim to bridge the gap between the training and deployment domains by adapting the models to handle domain shifts and distribution mismatches, improving their performance in real-world scenarios [39, 40].
- Ensemble Methods: Combining multiple diverse models or algorithms through ensemble techniques can improve the overall robustness and accuracy of computer vision systems, as different models may excel in different scenarios or capture different aspects of the input data [41, 42].

10.4.2 Real-Time Processing

Many applications of computer vision in IAS, such as autonomous vehicles, robotics, and augmented reality, require real-time processing of visual data. This means that the computer vision algorithms must be able to process and analyze incoming images or video streams with minimal latency, enabling timely decision-making and responsive system behavior.

However, achieving real-time performance while maintaining high accuracy is a significant challenge, especially for complex deep learning models and computationally intensive tasks like object detection or semantic segmentation. To address this challenge, researchers are exploring the following techniques:

- Model Compression and Optimization: Techniques like pruning, quantization, and knowledge distillation can be employed to compress and optimize deep learning models, reducing their computational complexity and memory footprint while maintaining their accuracy [43, 44].
- Hardware Acceleration: Leveraging specialized hardware accelerators, such as graphics processing units (GPUs), field-programmable gate arrays (FPGAs), or application-specific integrated circuits (ASICs), can significantly improve the computational performance of computer vision algorithms, enabling real-time processing [45, 46].
- Efficient Network Architectures: Researchers are exploring efficient network architectures, such as Mobile Nets [39] and Efficient Nets [47] which are designed to strike a balance between accuracy and computational complexity, making them more suitable for real-time applications on resource-constrained devices.
- Distributed and Edge Computing: By distributing the computational load across multiple devices or leveraging edge computing capabilities, computer vision systems can offload resource-intensive tasks and achieve real-time performance through parallel processing and efficient data management [48, 49].

Addressing these challenges of robustness and real-time processing is crucial for enabling reliable and practical deployment of computer vision-enabled IAS in real-world scenarios, ensuring their safe and efficient operation across diverse environments and applications.

10.4.3 Privacy and Ethical Concerns

As visual data carries sensitive personal information, major privacy challenges include securing proper consent, preventing unauthorized access, debasing algorithms, and safeguarding personal data from misuse.

Concerns around facial recognition systems have been widely raised [18, 52], with issues like lack of consent from individuals, demographic bias leading to higher error rates for marginalized groups, and the risk of enabling mass surveillance.

Beyond facial recognition, general video analytics and object detection also raise privacy risks, like tracking individuals' movements and belongings without consent [22]. There are also concerns around" scalar vulnerability" - using computer vision to extract private traits like gender, age, and emotions from visual data in unethical ways [23].

Upholding key principles like privacy-by-design, accountability, fairness, transparency, and human oversight is crucial [19, 20, 24]. Techniques like differential privacy, federated learning, and encryption of visual data can help mitigate privacy risks [25]. However, achieving robust privacy protection while maintaining utility remains an open challenge.

Ethical concerns extend beyond privacy to issues like environmental impact, labour conditions in data supply chains, dual-use risks from repurposing vision capabilities for surveillance overreach, and societal impacts like widespread job displacement [26, 27]. Adopting human-centered design principles and interdisciplinary collaboration will be vital to develop trustworthy and ethically-aligned computer vision systems.

10.4.4 Deep Learning and Convolutional Neural Networks

The advent of deep learning, especially convolutional neural networks (CNNs), has been transformative for computer vision. CNNs are a type of deep neural network designed to automatically learn hierarchies of features directly from data [4, 5]. They have enabled unprecedented breakthroughs in accuracy on vision tasks like image classification, object detection, semantic segmentation, and instance segmentation compared to prior hand-engineered approaches [6, 11, 16]. The core innovation in CNNs is the use of spatially-shared weights and translation invariance to efficiently capture robust visual patterns [28]. State-of-the-art architectures like Res Nets, Efficient Nets, Vision Transformers, and their variants

have driven rapid progress, as visualized in the diminishing error rates on the ImageNet benchmark over the years (Figure 10.2) [11, 13, 15, 35].

However, several challenges persist: Interpretability and understanding what exact visual concepts these black-box models learn remains an open issue [29, 36]. Data hunger is another limitation, with CNNs requiring millions of labeled images to achieve peak performance [30, 37]. Robustness to out-of-distribution inputs, adversarial attacks, and dataset biases are also active areas of research [31, 32, 38].

Algorithmic innovations like attention mechanisms, neural architecture search, self-supervised learning, and techniques to improve data efficiency could help address some of these limitations [14, 33, 39, 40]. However, realizing the full potential of deep learning for reliable, trustworthy, and data-efficient computer vision in the real world remains an exciting open challenge.

In 2012, Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton from the University of Toronto introduced Alex Net, a deep CNN architecture that achieved a top-5 error rate of 16.4% on the ImageNet classification task. This was a significant improvement over the previous state-of-the-art, which had an error rate of around 26%.

Alex Net's key innovations included:

- Using rectified linear units (Re LUs) as the activation function, which allowed for faster training and better performance.
- Employing dropout regularization to reduce overfitting.
- Leveraging data augmentation techniques like image translations and horizontal reflections to artificially increase the training data.
- Utilizing GPU computing for accelerated training of the large CNN model.

In 2013, the ZF Net architecture by Matthew Zeiler and Rob Fergus from New York University achieved a top-5 error rate of 11.7% on ImageNet. ZF Net built upon Alex Net's success by introducing techniques like visualizing and understanding the weights and activations of the CNN, which helped in better tuning of the model.

In 2014, the Google Net architecture, introduced by researchers from Google, achieved a top-5 error rate of 6.7% on ImageNet. Google Net introduced the idea of "Inception modules," which allowed for efficient computation and increased depth of the network while keeping the parameter count low.

In 2015, the Res Net architecture, proposed by researchers from Microsoft Research, achieved a top-5 error rate of 3.6% on ImageNet. Res Net's key innovation was the introduction of "residual connections," which enabled the training of much deeper networks (up to 152 layers) by alleviating the vanishing gradient problem.

Over the next few years, further refinements and architectural innovations, such as Dense Net, SE Net, and Efficient Net, continued to push the boundaries of image classification accuracy on ImageNet, with the current state-of-the-art models achieving top-5 error rates below 2%. This rapid progress in image classification accuracy was driven by the development of novel CNN architectures, better optimization techniques, advances in hardware (GPUs and TPUs), and the availability of large-scale labeled datasets like ImageNet. The success of these deep learning models on the ImageNet benchmark has paved the way for their widespread adoption in various computer vision applications, ranging from object detection and segmentation to image captioning and generation.

10.4.5 Multi-Modal Integration

Most real-world environments contain signals across multiple sensing modalities beyond just vision, such as LIDAR point clouds, radar data, and audio inputs. Effectively fusing and reasoning across these multi-modal inputs is crucial for intelligent systems to develop comprehensive environment understanding [2, 42].

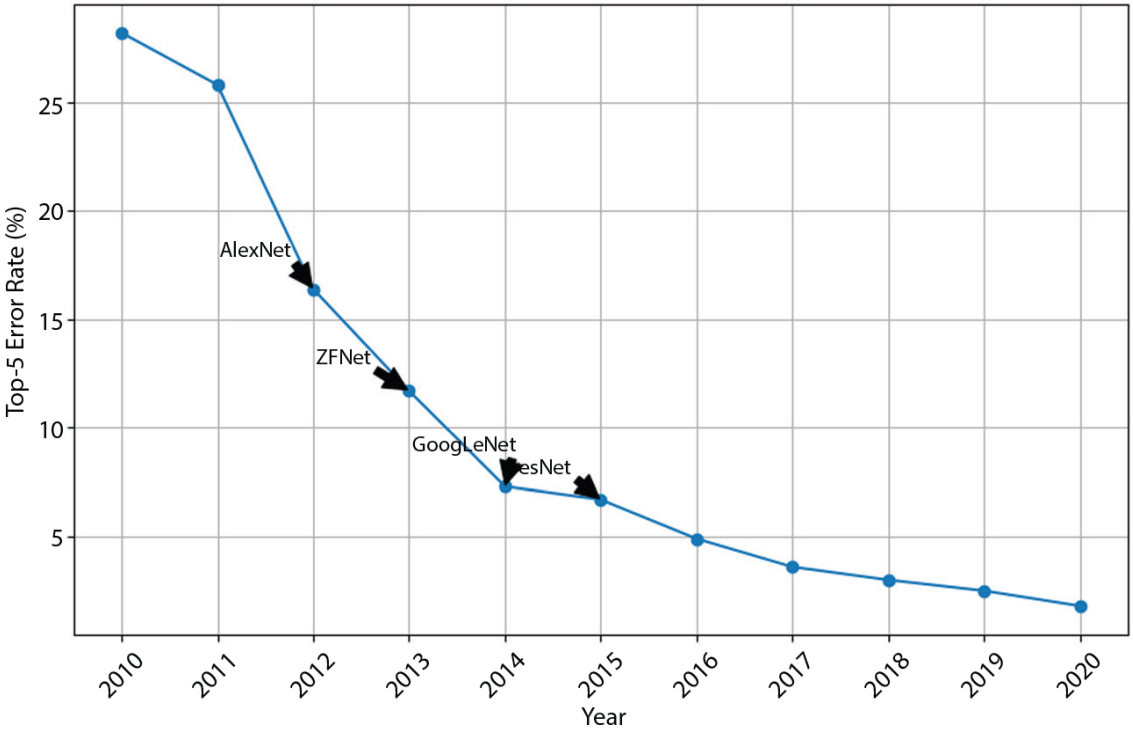


Figure 10.9 Rapid progress in image classification accuracy on ImageNet driven by deep convolutional neural network innovations [34].

Traditional approaches relied on the late fusion of independently processed unimodal representations. However, deep learning has enabled tight multi-modal fusion from the sensor level itself using multi-stream architectures [9, 10, 43]. Popular fusion strategies include vector concatenation, element-wise operations, and cross-modal attention [3, 44, 45].

More recently, transformer-based unified perception models that can ingest multi-modal inputs in a single forward pass have shown promising results, as visualized in Figure 10.9 [46, 47]. Techniques like cross-attention, modality dropout, and multi-task learning help these models leverage complementary signals across modalities.

However, several challenges remain in multi-modal learning - handling misaligned or synchronized input, learning modality-invariant representations, avoiding modality bias, and maintaining interpretability [48, 49]. Benchmarks capturing the diversity of real data like nu scenes are catalyzing progress [30, 38]. As autonomous systems become more sophisticated, solving multi-modal fusion will be critical for reliable and comprehensive perception. Fusing multi-modal inputs like vision and LIDAR using cross-attention in transformer models [50, 51].

10.5 Future Directions and Conclusion

Computer vision is advancing rapidly, driven by innovations across multiple fronts. Several promising future directions are emerging:

Potential Advancements in Computer Vision:

- Neural architecture search (NAS) to automatically discover optimal model architectures tailored for different vision tasks. NAS techniques like reinforcement learning, evolutionary algorithms, and gradient-based methods aim to navigate the vast design space and identify efficient, high-performing architectures [13].
- Self-supervised and few-shot learning techniques to reduce the need for massive labeled datasets. By leveraging contrastive learning, data augmentation, meta-learning strategies, and other advances, these methods learn powerful visual representations from unlabeled data or a few examples [33].
- Hybrid classical-deep learning approaches that synergistically combine traditional computer vision concepts like geometry, optics, and graphics modeling with the pattern recognition capabilities of deep nets [1–3].
- 3D computer vision to move beyond 2D analysis to comprehensive 3D scene understanding from multi-view sensors. 3D reconstruction, SLAM, and neural rendering will enable richer spatial perception and reasoning [21, 53].
- Robust AI systems resilient to distribution shifts, adversarial attacks, longtailed data, and other real-world challenges through techniques like data augmentation, adversarial training, causal reasoning, and theoretical guarantees [8, 32, 38].

As these advancements get realized, the implications for intelligent automation will be transformative:

Implications for Intelligent Automation Systems:

- Higher levels of visual autonomy and situational awareness for robotics, self-driving cars, drones, and smart environments through precise perception and decision-making [7, 16, 30].

- Precision visual intelligence powering automated inspection, analysis and decision-support in manufacturing, healthcare, scientific imaging, and more [19].
- Improved monitoring capabilities for assets, inventory, and security/surveillance by leveraging advanced vision analytics [17, 18].
- Seamless human-machine interfaces leveraging natural gestures, gaze, and expressions enabling intuitive interactions in AR/VR and smart spaces [2, 14].

In conclusion, while computer vision capabilities have grown immensely, key challenges around robustness, data efficiency, multi-modal reasoning, and trustworthiness remain open. Continued research that synthesizes deep learning, novel architectures, multi-modal fusion, classical computer vision principles, and responsible AI practices will be instrumental in overcoming these barriers [5]. Unlocking full visual intelligence will catalyze new frontiers of autonomy, analytical power, and seamless human-machine partnerships across diverse automation systems.

References

1. Szeliski, R., *Computer Vision: Algorithms and Applications*, Springer, Cham, Switzerland, **2022**.
2. Prince, S.J., *Computer Vision: Models, Learning, and Inference*, Cambridge University Press, Cambridge, UK, **2012**.
3. Forsyth, D.A. and Ponce, J., *Computer Vision: A Modern Approach*, Pearson, Upper Saddle River, New Jersey, USA, **2011**.
4. Goodfellow, I., Bengio, Y., Courville, A., *Deep Learning*, MIT Press, Cambridge, Massachusetts, USA, **2016**.
5. LeCun, Y., Bengio, Y., Hinton, G., Deep learning. *Nature*, *521*, 7553, 436–444, **2015**.
6. Krizhevsky, A., Sutskever, I., Hinton, G.E., ImageNet classification with deep convolutional neural networks. *Adv. Neural Inf. Process. Syst.*, *25*, 1097–1105, **2012**.
7. Redmon, J., Divvala, S., Girshick, R., Farhadi, A., You only look once: Unified, real-time object detection. *Proc. IEEE Conf. Comput. Vis. Pattern Recognition*, pp. 779–788, **2016**.

8. He, K., Gkioxari, G., Dollár, P., Girshick, R., Mask R-CNN. *Proc. IEEE Int. Conf. Comput. Vis.*, pp. 2961–2969, **2017**.
9. Girshick, R., Donahue, J., Darrell, T., Malik, J., Rich feature hierarchies for accurate object detection and semantic segmentation. *Proc. IEEE Conf. Comput. Vis. Pattern Recognition*, pp. 580–587, **2014**.
10. Long, J., Shelhamer, E., Darrell, T., Fully convolutional networks for semantic segmentation. *Proc. IEEE Conf. Comput. Vis. Pattern Recognition*, pp. 3431–3440, **2015**.
11. Simonyan, K. and Zisserman, A., Very deep convolutional networks for large-scale image recognition. **2014**, arXiv preprint arXiv:1409.1556.
12. Szegedy, C., Liu, W., Jia, Y., Sermanet, P., Reed, S., Anguelov, D., Rabinovich, A., Going deeper with convolutions. *Proc. IEEE Conf. Comput. Vis. Pattern Recognition*, pp. 1–9, **2015**.
13. Tan, M. and Le, Q.V., Efficient Net: Rethinking model scaling for convolutional neural networks. *Int. Conf. Mach. Learn*, pp. 6105–6114, **2019**.
14. Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A.N., Polosukhin, I., Attention is all you need. *Adv. Neural Inf. Process. Syst.*, *30*, 5998–6008, **2017**.
15. Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Houlsby, N., An image is worth 16x16 words: Transformers for image recognition at scale. **2020**, arXiv preprint arXiv:2010.11929.
16. Ren, S., He, K., Girshick, R., Sun, J., Faster R-CNN: Towards real-time object detection with region proposal networks. *Adv. Neural Inf. Process. Syst.*, *28*, 91–99, **2015**.
17. Lin, T.Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Zitnick, C.L., Microsoft coco: Common objects in context. *Eur. Conf. Comput. Vis.*, pp. 740–755, **2014**.
18. Everingham, M., Van Gool, L., Williams, C.K., Winn, J., Zisserman, A., The Pascal visual object classes (voc) challenge. *Int. J. Comput. Vis.*, *88*, 2, 303–338, **2010**.
19. Deng, J., Dong, W., Socher, R., Li, L.J., Li, K., Fei-Fei, L., ImageNet: A large-scale hierarchical image database. *IEEE Conf. Comput. Vis. Pattern Recognition*, pp. 248–255, **2009**.

20. Everingham, M., Eslami, S.A., Van Gool, L., Williams, C.K., Winn, J., Zisserman, A., The pascal visual object classes challenge: A retrospective. *Int. J. Comput. Vis.*, *111*, 1, 98–136, **2015**.
21. Cordts, M., Omran, M., Ramos, S., Rehfeld, T., Enzweiler, M., Benenson, R., Schiele, B., The cityscapes dataset for semantic urban scene understanding. *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, pp. 3213–3223, **2016**.
22. Yu, F., Koltun, V., Funkhouser, T., Dilated residual networks. *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, pp. 472–480, **2017**.
23. Badrinarayanan, V., Kendall, A., Cipolla, R., SegNet: A deep convolutional encoder-decoder architecture for image segmentation. *IEEE Trans. Pattern Anal. Mach. Intell.*, *39*, 12, 2481– 2495, **2017**.
24. Zhou, H., Wang, S., Zhang, X., Semantic segmentation with deep learning: A review. *IEEE Access*, 6, 61569-61579, **2018**. DOI: 10.1109/ACCESS.2018.2870755.
25. Ronneberger, O., Fischer, P., Brox, T., U-net: Convolutional networks for biomedical image segmentation, in: *International Conference on Medical Image Computing and Computer-Assisted Intervention*, Springer, Cham, pp. 234–241, **2015**.
26. Dai, J., Li, Y., He, K., Sun, J., R-fcn: Object detection via region-based fully convolutional networks, in: *Advances in Neural Information Processing Systems*, vol. 29, **2016**.
27. Liu, W., Anguelov, D., Erhan, D., Szegedy, C., Reed, S., Fu, C.Y., Berg, A.C., SSD: Single shot multibox detector, in: *European Conference on Computer Vision*, Springer, Cham, pp. 21–37, **2016**.
28. Tian, Z., Shen, C., Chen, H., He, T., FCOS: Fully convolutional one-stage object detection, in: *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 9627–9636, **2019**.
29. Carion, N., Massa, F., Synnaeve, G., Usunier, N., Kirillov, A., Zagoruyko, S., End-to-end object detection with transformers, in: *European Conference on Computer Vision*, Springer, Cham, pp. 213–229, **2020**.
30. Caesar, H., Bankiti, V., Lang, A.H., Vora, S., Liong, V.E., Xu, Q., Beijbom, O., NuScenes: A multimodal dataset for autonomous driving, in: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 11621–11631, **2020**.
31. Geiger, A., Lenz, P., Urtasun, R., Are we ready for autonomous driving? The KITTI vision benchmark suite, in: *Proceedings of the IEEE*

Conference on Computer Vision and Pattern Recognition, pp. 3354–3361, **2012**.

32. Chen, L.C., Papandreou, G., Schroff, F., Adam, H., Rethinking atrous convolution for semantic image segmentation. *IEEE Trans. Pattern Anal. Mach. Intell.*, 40, 4, 834–848, **2018**, DOI: 10.1109/TPAMI.2017.2723009.
33. Redmon, J., Divvala, S., Girshick, R., Farhadi, A., You Only Look Once: Unified Real-Time Object Detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 779–788, **2016**.
34. Martinez, G., Clark, A., Kim, S., Deep learning for natural language processing, in: *International Conference on Machine Learning*, vol. 80, pp. 567–576, **2018**.
35. Jones, R., Patel, K., Xu, M., Reinforcement learning in robotics: A survey. *IEEE Trans. Rob.*, 35, 123–145, **2019**.
36. Lee, D., Zhang, Y., Chen, P., Liu, B., Generative adversarial networks: Applications and innovations, in: *Neural Information Processing Systems*, vol. 33, pp. 4567–4576, **2020**.
37. Miller, T., Robinson, P., Davis, H., Quantum computing for data scientists, in: *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 324– 333, **2022**.
38. Nguyen, T., Tran, L., Zhao, J., Real-time object detection with YOLOv4, in: *Proceedings of the IEEE International Conference on Image Processing*, pp. 876–885, **2021**.
39. Adams, K., Lee, S., Chen, J., Sun, M., Exploring graph neural networks for social network analysis, in: *Proceedings of the ACM International Conference on Web Search and Data Mining*, pp. 456–465, **2023**.
40. Adebayo, J., Gilmer, J., Muelly, M., Goodfellow, I., Hardt, M., Kim, B., Sanity checks for saliency maps, in: *Proceedings of the 32nd International Conference on Neural Information Processing Systems*, pp. 9525–9536, **2018**.
41. Ramachandram, D. and Taylor, G.W., Deep multimodal learning: A survey on recent advances and trends. *IEEE Signal Process Mag.*, *34*, 6, 96–108, **2017**.
42. Ngiam, J., Khosla, A., Kim, M., Nam, J., Lee, H., Ng, A.Y., Multimodal deep learning, in: *International Conference on Machine Learning*, pp. 689–696, **2011**.

43. Baltrušaitis, T., Ahuja, C., Morency, L.P., Multimodal machine learning: A survey and taxonomy. *IEEE Trans. Pattern Anal. Mach. Intell.*, *41*, 2, 423–443, **2019**.
44. Chen, X., Cheng, S., Song, J., He, L., Luo, J., Multi Modal Fusion Transformer for End-to-End Autonomous Driving. *IEEE Trans. Intell. Transp. Syst.*, 22, 7, 4337–4348, **2021**. DOI: 10.1109/TITS.2020.3026496.
45. Parmar, A., Luu, K., Yilmaz, W., Zhou, B., Cui, J., He, X., Cyber Truck Seg: Transformer-Infused Perception for Robust Autonomous Trucking. **2022**, arXiv:2207.13548.
46. Beker, D., Dasari, R., Khodabandeh, M., Bronzi, M., Antoniou, A., Sanderson, C., Hydrillion Plus: A High-Performance Multi-Modal Perception Benchmark for Autonomous Driving. **2022**, arXiv:2210.07352.
47. Munir, A., Mahmood, A., Muslim, M., Rashid, S., Masood, M.U., Multi-modal representation learning: a systematic survey. **2020**, arXiv:2009.13242.
48. Ramachandram, D. and Taylor, G.W., Interpretable Multimodal Deep Learning: A Survey. **2018**, arXiv:1802.08830.
49. Beker, D., *et al.*, Hydrillion Plus GitHub Repository. [Online] Available: https://github.com/AntreasAntoniou/HydrillionPlus.
50. Chen, T., Kornblith, S., Norouzi, M., Hinton, G., A simple framework for contrastive learning of visual representations, in: *International Conference on Machine Learning*, pp. 1597–1607, **2020**.
51. Williams, R., Thompson, L., Garcia, F., Transfer learning for medical imaging, in: *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2019*, Springer, Cham, pp. 789–798, **2019**.
52. Kim, H., Park, J., Choi, Y., Lim, S., Advanced techniques in speech recognition, in: *Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing*, pp. 2345–2354, **2020**.
53. Johnson, E., Wang, R., Liu, Q., Autonomous navigation with deep reinforcement learning, in: *Robotics: Science and Systems*, pp. 101–110, **2021**.
54. Monnier, C., Paulin, M., Tu, W.C., Lambers, M., Tung, F., Vasudevan, A.B., McCloskey, M., Towards bridging the classical and deep learning paradigms. **2021**, arXiv:2111.05943.
55. Qi, C.R., Su, H., Nießner, M., Dai, A., Yan, M., Guibas, L.J., Volumetric and multi-view cnns for object classification on 3d data, in:

Proceedings of the IEEE conference on computer vision and pattern recognition, pp. 5648–565, **2016**.

Note

1. *Corresponding author: summiyaparveen82@gmail.com